PHYSWEEP: Does a Video Generator Realize the Physics You Ask For?

Supplementary Material

S1. Measurement pipeline and failure-form estimation

Algorithm 1 gives the full measurement loop referenced from the main paper's Suite, Pipeline, and Protocol section.

Algorithm 1 PHYSWEEP measurement, one system and one model

Require: frozen generator G; sweep grid $\{\theta_i\}$; seeds $\{z_j\}$; law f; tracker T; fit-quality gate ρ; VLM rater P

for each θ_i and seed z_j **do**
  $c_{ij} \leftarrow \text{RENDER}(s_0(z_j), \theta_i)$
  $v_{ij} \leftarrow G(c_{ij})$ ▷ inference only, weights frozen
  $o_{ij} \leftarrow T(v_{ij})$ ▷ blob centroid or CoTracker3
  $(\hat{\theta}_{ij}, R^2_{ij}) \leftarrow \text{FIT}(f, o_{ij})$
  if $R^2_{ij} < \rho$ **then**
    mark v_{ij} **non-physical**; exclude from PRE
  end if
  $p_{ij} \leftarrow P(v_{ij})$ ▷ plausibility contrast
end for
compute PRE, slope β, Spearman ρ_s, EG, PRI, CBI
bootstrap 95% CIs over (θ_i, seed) pairs
return metrics, gate-dropped rate, plausibility pairs

H1 (shrinkage, PRI**).** Given pairs $(\theta_i, \hat{\theta}_{ij})$, fit $\hat{\theta} = \alpha\theta + (1-\alpha)\theta_0$ by least squares with θ_0 either fixed to the physically typical value or estimated jointly; $\text{PRI} = 1 - \hat{\alpha}$, clipped to $[0, 1]$.

H2 (clamp, CBI**).** Fit $\hat{\theta} = \min(\theta, \theta_{\text{edge}})$ with θ_{edge} at the in-range boundary or estimated; $\text{CBI} = 1 - \text{RSS}_{\text{H2}}/\text{RSS}_{\text{null}}$.

Selection. Compare H1 and H2 by leave-one-θ-out R^2 and by ΔAIC. A hypothesis is selected only if its held-out R^2 is positive; if both fail that test the pair is reported as *neither* and the observed behavior is described without forcing a label. This is what happens on all eight adjudicated pairs in the main paper, for the reason given there. All confidence intervals use the paired bootstrap over (θ_i, seed).

When CBI **is not identifiable.** The released smoke test records a limitation of CBI as defined. If an out-of-range grid lies entirely on one side of the in-range band, the clamp prediction $\min(\theta, \theta_{\text{edge}})$ is constant across S_{out} and is therefore indistinguishable from the null mean model, so CBI ≈ 0 and clamping cannot be adjudicated at any sample size. Five of our six axes use two-sided out-of-range grids (Table 1) and are unaffected. The exception is pendulum damping, whose out-of-range grid $\{0.3, 0.6, 0.9\}$ lies entirely above the in-range grid; no result in the main paper depends on this, because every pendulum row is gate-dropped at 96.7–100% and carries no adjudication. A two-sided damping grid would be needed to test clamping on that axis.

S2. Consistency of the recovery estimator: full statement and proof

We restate the main paper's consistency proposition with explicit regularity conditions and give the argument in full. The claim concerns only the estimator R, not the generator G.

Setup. Fix a system with law f and scalar parameter $\theta \in \Theta$, with Θ a compact interval. A clip of T frames at a given resolution yields a tracked observable $o \in \mathbb{R}^d$ (object centers, angles, or contact times). Write the population fitting statistic as $\mathcal{T}(\theta) \in \mathbb{R}^k$: for the projectile, the quadratic coefficient of height against time; for the pendulum, the mean zero-crossing interval; for the bouncing ball, the successive height ratio. The estimator forms an empirical statistic $\hat{\mathcal{T}}$ from o and returns $\hat{\theta} = g(\hat{\mathcal{T}})$, where $g = \mathcal{T}^{-1}$ on the range of $\mathcal{T}$.

Assumptions. **(A1, trackability)** the tracker returns $\hat{o} = o^\star + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2\Sigma$ for a fixed positive-definite Σ, and $\sigma^2 \to 0$ as T and the resolution grow. **(A2, identifiability)** $\mathcal{T} : \Theta \to \mathcal{T}(\Theta)$ is injective and continuously differentiable with $\|\mathcal{T}'(\theta)\| \geq c > 0$ on Θ. **(A3, regularity)** the empirical statistic is a continuous functional $\hat{\mathcal{T}} = \Phi(\hat{o})$ with Φ continuous at $o^\star$ and $\Phi(o^\star) = \mathcal{T}(\theta)$.

Proposition (Consistency of R, restating the main paper)**.** *Under (A1)–(A3),* $\hat{\theta} = g(\Phi(\hat{o})) \xrightarrow{p} \theta$ *as* $\sigma^2 \to 0$.

Proof. By (A2), $\mathcal{T}$ is a continuous injection on the compact set Θ, so its inverse $g = \mathcal{T}^{-1}$ is continuous on $\mathcal{T}(\Theta)$: a continuous bijection from a compact space to a Hausdorff space is a homeomorphism. By (A1), $\text{Var}(\hat{o}) = \sigma^2\Sigma \to 0$ with mean $o^\star$, so $\hat{o} \to o^\star$ in mean square and hence in probability. By (A3), Φ is continuous at $o^\star$, so the continuous mapping theorem gives $\hat{\mathcal{T}} = \Phi(\hat{o}) \xrightarrow{p} \Phi(o^\star) = \mathcal{T}(\theta)$. Ap-

plying the continuous mapping theorem again with the continuous g yields $\hat{\theta} = g(\hat{\mathcal{T}}) \xrightarrow{p} g(\mathcal{T}(\theta)) = \theta$. □

Per-system check of (A2)–(A3). *Projectile.* $\mathcal{T}(\theta) = -\theta/2$ is linear, hence injective with $\mathcal{T}' = -1/2$, and Φ is ordinary least squares on a quadratic basis, continuous in the data.

Pendulum, and a bias this exposes. For the undamped case $\zeta = 0$, $\mathcal{T}(\omega) = 2\pi/\omega$ on $\omega > 0$ is strictly monotone with nonvanishing derivative on any Θ bounded away from 0, and zero-crossing interpolation is continuous where crossings are simple. Our pendulum is damped ($\zeta > 0$ throughout, Section S4), so the statistic realized by zero-crossing spacing is $\mathcal{T}(\omega, \zeta) = 2\pi/(\omega\sqrt{1-\zeta^2})$, the *damped* frequency ω_d, not the natural frequency ω. The map remains injective in ω for fixed ζ, so Proposition S2 still guarantees consistent recovery of ω_d; the reported $\hat{\omega}$ is therefore a slightly biased estimate of ω whenever $\zeta > 0$. Verified numerically, the bias is $< 1.3\%$ at the in-range ζ grid $\{0.02, 0.05, 0.1\}$, grows with ζ, and is materially larger at the out-of-range grid $\{0.3, 0.6, 0.9\}$. We did not correct for it post hoc, since doing so would change every already-reported $\hat{\omega}$; we state it so the recovery is not mistaken for exact, and note it compounds with the heavily-damped, few-peaks low-fidelity regime for this axis. In practice this caveat has little effect on the reported results, because every pendulum row in the main paper is gate-dropped at 96.7–100% and carries no slope estimate.

Bouncing ball. The map $e \mapsto e^2$ from restitution to successive height ratio is injective and smooth on $(0, 1]$.

The fit-quality gate ($R^2 < \rho$) excludes generations on which Φ is ill-posed, for example when the object morphs or vanishes, so (A3) is enforced operationally rather than assumed.

S3. Seed-conditional defaults: full statement and proof

This section proves the main paper's Proposition on misspecification and records the estimator for the Seed Determination Index.

Model. The sweep is a *crossed* design: every grid value θ_i is run with every seed z_j, so the seed is independent of θ by construction. Under a seed-conditional default, $\hat{\theta}_{ij} = c(z_j) + \varepsilon_{ij}$ with $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma_\varepsilon^2$, $\varepsilon \perp (\theta, z)$, and $\sigma_c^2 = \mathrm{Var}_z[c(z)] > 0$. The crossing assumption is load bearing and we flag it for anyone reusing the protocol. If seeds were confounded with grid position, so that c varied systematically with θ, statement (i) fails and the failure is not subtle: in simulation a generator whose output is entirely seed-determined, with no response to θ whatsoever, returns $\hat{\alpha} = 1.006$ and $\mathrm{PRI} = -0.006$ under a confounded design, which reads as *perfect faithfulness*. Crossing seeds with grid values is therefore a correctness requirement of the measurement, not a convenience.

Proposition (Restating Proposition 1 of the main paper)**.** *Under this model: (i) the pooled OLS fit of $\hat{\theta} = \alpha\theta + b$ satisfies $\hat{\alpha} \xrightarrow{p} 0$, so* $\mathrm{PRI} \to 1$*; (ii) $\hat{b} \xrightarrow{p} \mathbb{E}_z[c(z)]$ and the implied $\hat{\theta}_0 = \hat{b}/(1-\hat{\alpha}) \xrightarrow{p} \mathbb{E}_z[c(z)]$; (iii) any measurable $h(\theta)$ has population $R^2 = 0$ for predicting $\hat{\theta}$, its leave-one-θ-out estimate satisfies $R^2_{\mathrm{LOO}} \xrightarrow{p} 0$, and for any fixed $\tau > 0$ a rule accepting only $R^2_{\mathrm{LOO}} > \tau$ returns* neither *with probability tending to one. None of (i)–(iii) uses $\sigma_c^2 > 0$, so all three hold verbatim when $\sigma_c^2 = 0$ and σ_ε^2 is large.*

Proof. (i) $\mathrm{Cov}(\theta, \hat{\theta}) = \mathrm{Cov}(\theta, c(z)) + \mathrm{Cov}(\theta, \varepsilon) = 0$: the first term vanishes because the crossed design makes z independent of θ, the second because $\varepsilon \perp \theta$. The OLS slope is $\widehat{\mathrm{Cov}}(\theta, \hat{\theta})/\widehat{\mathrm{Var}}(\theta)$, which converges in probability to $0/\mathrm{Var}(\theta) = 0$ by the weak law, with $\mathrm{Var}(\theta) > 0$ since the grid is non-degenerate. Hence $\mathrm{PRI} = 1 - \hat{\alpha} \to 1$.

(ii) $\hat{b} = \bar{\hat{\theta}} - \hat{\alpha}\bar{\theta} \xrightarrow{p} \mathbb{E}[\hat{\theta}] = \mathbb{E}_z[c(z)]$, and dividing by $1 - \hat{\alpha} \to 1$ gives the same limit. This limit is a population average: when c takes two well-separated values it need not lie near either, so the fitted "default" describes no individual generation.

(iii) For any measurable h, $\mathbb{E}[(\hat{\theta} - h(\theta))^2] = \mathrm{Var}(\hat{\theta}) + \mathbb{E}[(\mathbb{E}[\hat{\theta}] - h(\theta))^2]$, the cross term vanishing by $\hat{\theta} \perp \theta$. This is minimized at $h \equiv \mathbb{E}[\hat{\theta}]$ with value $\mathrm{Var}(\hat{\theta})$, so the population R^2 is 0. The leave-one-θ-out numerator and denominator are averages of i.i.d. terms converging in probability to $\mathrm{Var}(\hat{\theta}) > 0$, so their ratio converges to 1 and $R^2_{\mathrm{LOO}} \xrightarrow{p} 0$ by the continuous mapping theorem. Consequently $\Pr[R^2_{\mathrm{LOO}} > \tau] \to 0$ for any fixed $\tau > 0$, which is the claim. Finally, no step above invokes $\sigma_c^2 > 0$; the argument uses only $\hat{\theta} \perp \theta$, which also holds when c is constant and the variance is carried entirely by ε. □

Remark 1 (What is deliberately not claimed)**.** *We do* not *claim $\mathbb{E}[R^2_{\mathrm{LOO}}] < 0$. Simulation under this model gives a mean of -0.024 at a 5×5 design shrinking to -0.001 at 20×20, so the expectation does appear non-positive and to vanish with sample size, but R^2_{LOO} is a ratio of dependent random variables and we have not discharged the argument rigorously. Nothing in the paper depends on it: convergence in probability to 0, together with the fixed margin, is what statement (iii) and the released selection rule actually use. We record it here as an observation rather than a result.*

Remark 2 (Why a margin is required in the selection rule)**.** *Because R^2_{LOO} is centred on zero under this model, its* sign *is noise. An acceptance test of the form $R^2_{LOO} > 0$ therefore labels unstructured data as prior reversion with probability approaching $1/2$. The released* `select_mechanism` *requires $R^2_{LOO} > 0.1$; the smoke*

test contains a synthetic sample on which the bare sign test returns `prior_reversion` *($R^2_{LOO} = 0.035$) while the margin rule correctly returns* `neither`*, and asserts both, so the margin cannot be removed silently.*

Estimating SDI**.** With k seeds, N gated clips and mean group size $n_0 = N/k$, let MS_b and MS_w be the between- and within-seed mean squares of $\hat{\theta}$. We report the one-way random-effects estimate

$$\mathrm{SDI} = \frac{\mathrm{MS}_b - \mathrm{MS}_w}{\mathrm{MS}_b + (n_0 - 1)\,\mathrm{MS}_w},$$

clipped to $[0, 1]$, which is consistent for $\sigma_c^2/(\sigma_c^2 + \sigma_\varepsilon^2)$. The uncorrected variance share is strongly upward-biased at this design size, returning about 0.18 in simulation where the truth is 0, which is precisely the regime in which a false positive would be most damaging; the corrected estimate returns about 0.07 there. FDI is the analogous share for the grid factor. Both return `NaN`, never a number, when fewer than two seeds are present, when $\hat{\theta}$ is constant, or when there is no within-seed replication.

S4. System grids, seed counts, and rendering

All scenes are rendered by a seeded deterministic 2D engine at 256×256 over $T{=}24$ frames at 24 fps, with a single high-contrast disk (radius 8 px) on a plain background and a fixed ground line. Only the swept parameter and the seeded initial condition vary within a system. Spatial and temporal scales are held fixed within a clip and all metrics use the ratio $\hat{\theta}/\theta$, so the result is invariant to a common rescaling. The in-range split is centered on values common in natural video; the out-of-range split probes rare or absent values. These are the grids fixed in the released config and used for every reported result.

System	θ	In-range grid	Out-of-range grid
Projectile	g	$\{5, 7, 9.8, 12, 15\}$	$\{1.6,\ 3,\ 20,\ 25\}$
Damped pendulum	ω	$\{2, 3, 4, 5\}$ rad/s	$\{0.5,\ 1,\ 8,\ 12\}$
	ζ	$\{0.02, 0.05, 0.1\}$	$\{0.3,\ 0.6,\ 0.9\}$
Bouncing ball	e	$\{0.6, 0.7, 0.8, 0.9\}$	$\{0.2,\ 0.35,\ 0.97\}$
Spring–mass	k	$\{20, 40, 60, 80\}$ N/m	$\{5,\ 10,\ 200,\ 400\}$
Inclined slide	μ	$\{0.1, 0.2, 0.3, 0.4\}$	$\{0.01,\ 0.7,\ 1.0\}$

Table 1. Sweep grids per system. Each row has one swept axis; $m \geq 5$ seeded initial conditions per grid point.

Seed counts. The main paper refers to this section for the exact seed counts behind each reported cell, including the targeted follow-up batch run on the two model-axis pairs closest to `select_mechanism`'s four-point leave-one-out threshold.

Model	Axis	seeds/pt	clips gen.	clips gated
[MISSING: fill this table from the raw result files: one row]				
[MISSING: per model-axis pair (18 rows), giving seeds per]				
[MISSING: grid point, total clips generated, and total clips]				
[MISSING: clearing the $R^2 \geq 0.8$ gate, split in/out-of-range]				

Table 2. Seed counts and clip yields per model-axis pair. [MISSING: also state separately, here, the size of the follow-up seed batch run on the two closest-to-threshold pairs, and the 110 out-of-range attempts cited for LTX-Video/pendulum-ω]

Models, prompts, and compute. [MISSING: exact model repository, revision/commit hash, weight version, and licence for LTX-Video, CogVideoX-5B-I2V, DynamiCrafter$_{512}$, CoTracker3, and Qwen2.5-VL-7B-Instruct, all pinned at run time]

[MISSING: the verbatim VLM plausibility prompt used for all 630 ratings, plus the two alternative generation prompt templates used in the prompt-phrasing ablation and the four classifier-free guidance values swept]

[MISSING: GPU model and VRAM, total wall-clock GPU hours, and approximate generation time per clip per model]

S5. Reproducibility checklist

WACV's author guide points to the Reproducibility Checklist as a guide rather than a mandated form, and encourages voluntary code submission. We record the substantive items here.

- **Code.** Metrics, bootstrap CIs, mechanism selection, and the synthetic smoke test are released; system generators are released. [MISSING: confirm the generation and tracking drivers are included too, or state explicitly which parts are not]
- **Data.** Fully procedural, no external assets, no human annotation, seeded and deterministic.
- **Models.** All public, all frozen, inference only. No training or fine-tuning was performed at any point.
- **Randomness.** Every generation is keyed to an explicit integer seed; $m \geq 5$ seeds per grid point; seeds are recorded per clip.
- **Statistics.** All intervals are 95% paired bootstraps over (θ_i, seed); effect sizes are reported alongside p-values; multiplicity across the six paired-bootstrap extrapolation tests is handled by a Holm correction.
- **Raw outputs.** [MISSING: state whether per-clip records (seed, θ, $\hat{\theta}$, fit R^2, gate decision, VLM score) are shipped with the supplementary code; if they are, name the file]